SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 2 – Agile Sprint Planning & User Story Workshop

Course Code:	CSA10 / CSA1005	**Course Title:**	Software Engineering
CO Assessed:	CO1	**Assessment:**	Assessment Tool 2 – **Agile Sprint Planning & User Story Workshop**
Weightage:	20%	**Total Marks:**	25
CO1 Statement:	*Apply advanced methodologies and agile project management techniques to manage software projects effectively*		
Student Name:	A.DEKKSHITHA	**Reg. No.:**	192572133
Date:	15.10.2026	**Duration:**	60 minutes

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Annexure A – Agile Sprint Planning & User Story Workshop

Instructions to Students:

Each student is assigned an individual software project problem statement. Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

1. **Identify the Project Vision and Stakeholders**
   - Define the project's vision, objectives, and identify all key stakeholders involved.
2. **Create Epics and User Stories**
   - Break down the requirements into Epics and formulate well-structured User Stories using the format: *As a <user>, I want <feature>, so that <benefit>.*
3. **Define Acceptance Criteria**
   - Write clear and measurable acceptance criteria for each selected user story using the Given–When–Then format or equivalent.
4. **Prioritize the Product Backlog**
   - Organize and prioritize the product backlog based on business value, customer needs, and project dependencies.
5. **Plan the Sprint Backlog**
   - Select high-priority user stories for the first sprint and divide them into actionable development tasks.
6. **Estimate Effort Using Story Points**
   - Assign story points to each selected user story using an appropriate estimation technique (e.g., Fibonacci sequence or Planning Poker) and justify your estimates.
7. **Present the Sprint Goal and Expected Deliverables**
   - Define a clear sprint goal and describe the expected sprint deliverables, including the features to be completed at the end of the sprint.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Project Vision & Stakeholder Identification (30%)	Project vision is clear, comprehensive, and aligned with the problem statement. All relevant stakeholders and their roles are correctly identified.	Vision is clear with minor omissions. Most stakeholders are correctly identified.	Vision is partially defined. Some stakeholders are missing or incorrectly identified.	Vision is unclear or incomplete. Stakeholders are poorly identified or missing.
Epics & User Stories (25%)	Epics are well-defined and user stories are complete, follow Agile format, and fully capture functional requirements.	Epics and user stories are mostly complete with minor formatting or requirement issues.	User stories are partially complete with limited Agile structure.	User stories are incomplete, unclear, or do not follow Agile practices.
Acceptance Criteria (20%)	Acceptance criteria are specific, measurable, testable, and aligned with all user stories.	Acceptance criteria are mostly clear and testable with minor gaps.	Acceptance criteria are partially defined and lack clarity or completeness.	Acceptance criteria are missing or not aligned with user stories.
Product Backlog Prioritization & Sprint Planning (15%)	Product backlog is logically prioritized, sprint backlog is realistic, and task breakdown is complete.	Backlog prioritization and sprint planning are mostly appropriate with minor issues.	Prioritization or sprint planning is partially complete with limited justification.	Backlog prioritization and sprint planning are poorly organized or incomplete.
Story Point Estimation, Sprint Goal & Presentation (10%)	Story point estimation is well justified, sprint goal is clear, deliverables are realistic, and presentation is professional and well organized.	Estimation and sprint goal are mostly appropriate with minor inconsistencies. Presentation is clear.	Estimation or sprint goal lacks justification. Presentation is adequate but incomplete.	Estimation is unrealistic or missing. Sprint goal, deliverables, and presentation are unclear or incomplete.

Criteria	Mark
Project Vision & Stakeholder Identification **(30%)**	/5
Epics & User Stories **(25%)**	/5
Acceptance Criteria **(20%)**	/5
Product Backlog Prioritization & Sprint Planning **(15%)**	/5
Story Point Estimation, Sprint Goal & Presentation **(10%)**	/5
TOTAL	/25

Smart Campus Lost-and-Found Management System

1. Project Vision and Stakeholders

Project Vision

To provide a centralized, secure, and easy-to-use campus platform where students and staff can report lost or found belongings, attach photographs, record locations and timestamps, identify possible matches, and coordinate verified item recovery. The system replaces scattered notices and manual records with searchable digital reports and a traceable handover process.

Objectives

- Allow users to submit lost-item and found-item reports with descriptions, category, photograph, location, date and time.
- Help users and administrators identify possible lost/found matches.
- Notify potential owners when a likely match is found.
- Verify ownership before handing over an item and maintain recovery history.
- Provide basic analytics on reports, matches and successful recoveries.

Key Stakeholders and Responsibilities

Stakeholder	Role / Interest
Students and staff	Report lost/found belongings, search listings, respond to matches and collect verified items.
Campus security / lost-and-found desk	Review reports, check item custody, verify ownership and authorize handovers.
System administrator	Manage accounts, categories, access permissions, suspicious reports and system settings.
Project team / developers	Gather requirements, design, implement, test and maintain the platform.
Campus management	Monitor recovery trends, review analytics and ensure campus procedures are followed.

Project boundary: The first release focuses on reporting, searching, matching, notifications, verification and recovery records within one campus. It does not guarantee automatic identification or replace staff verification.

2. Epics and User Stories

User stories are written from the perspective of a user and describe a feature together with its benefit. The stories below are grouped into epics that represent major areas of functionality.

Epic ID / Epic	Story ID	User Story
E1 – Item Reporting	US-01	As a student or staff member, I want to submit a lost-item report with a description, category, location and date/time, so that others can help identify my belonging.
E1 – Item Reporting	US-02	As a finder, I want to submit a found-item report and upload a photograph, so that the owner can recognize the item.
E2 – Search and Matching	US-03	As a user, I want to search and filter reports by item category, keywords, location and date, so that I can find relevant listings quickly.
E2 – Search and Matching	US-04	As a user, I want the system to suggest possible lost/found matches using shared details, so that potential recoveries can be identified.
E3 – Notifications and Recovery	US-05	As a potential owner, I want to receive a notification when a likely match is found, so that I can respond promptly.
E3 – Notifications and Recovery	US-06	As a campus lost-and-found staff member, I want to verify ownership before handover and record the result, so that items are returned securely.
E4 – Administration and Analytics	US-07	As an administrator, I want to review, update or flag reports, so that inaccurate or inappropriate listings can be managed.
E4 – Administration and Analytics	US-08	As campus management, I want to view counts of reports, matches and recovered items, so that I can evaluate the service.

Epic Completion / Business Value

- E1 makes reports consistent by collecting the key information required to identify an item.
- E2 reduces manual searching by making listings searchable and surfacing possible matches.
- E3 supports timely communication and safer item handover.
- E4 supports responsible system operation and helps management assess outcomes.

Definition of Done (applies to each story)

- The feature meets its acceptance criteria and has been reviewed.
- Required fields are validated and errors are explained clearly to the user.
- Relevant data is saved and can be retrieved correctly.
- Basic functional testing is completed; access to personal contact details is restricted to authorized users.

3. Acceptance Criteria

Acceptance criteria are measurable conditions used to confirm whether a user story is complete. The following Given–When–Then checks apply to the selected stories.

US-01: Submit a Lost-Item Report

- Given a signed-in user is on the report form, when they enter the required description, category, location and approximate date/time and submit, then the system saves the report and displays a unique report reference.
- Given a required field is missing or invalid, when the user submits the form, then the report is not saved and the missing or invalid field is identified.

US-02: Submit a Found-Item Report

- Given a finder enters the item description, category, found location and found date/time, when they submit the report, then the system saves it as a found-item listing.
- Given an optional photograph is selected in an allowed format and within the configured size limit, when the report is submitted, then the image is attached to the listing; invalid files are rejected with a clear message.

US-03: Search and Filter Reports

- Given reports exist, when a user searches by keyword or applies category, location or date filters, then only listings matching the selected search conditions are displayed.
- Given no report matches the search, when results are displayed, then the system shows a clear “No matching reports found” message and allows the user to change the filters.

US-04: Suggest Possible Matches

- Given a lost report and one or more found reports share relevant details, when matching is run, then the system ranks candidates using available fields such as category, keywords, location and date.
- Given a candidate is suggested, when the user or staff opens it, then the system labels it as a possible match and does not treat the match as proof of ownership.

US-05: Notify Potential Owner

- Given a possible match passes the configured matching threshold, when the match is created, then the system creates a notification for the relevant reporter using the contact channel configured for the prototype.
- Given notification delivery fails, when the system records the failure, then the match remains available in the application and the failure is logged for review.

US-06: Verify Ownership and Record Handover

- Given a claimant requests an item, when staff checks the claimant's distinguishing details or proof of ownership, then staff can record the verification outcome as approved or rejected.
- Given ownership is approved and the item is handed over, when staff records the handover date, recipient reference and staff member, then the report status changes to "Recovered/Returned" and the recovery history is saved.
- Given verification is not successful, when staff rejects the claim, then the item is not marked as returned and the reason is recorded for authorized staff.

US-07: Moderate Reports

- Given an administrator is authenticated with the correct role, when they review a report, then they can update its status or flag/remove inappropriate content.
- Given a user without administrator permission attempts a moderation action, when the action is requested, then access is denied and no report data is changed.

US-08: View Analytics

- Given report and recovery records exist, when an authorized administrator opens analytics, then the dashboard displays total lost reports, found reports, possible matches and recovered items.
- Given the date range is changed, when the dashboard refreshes, then the displayed counts reflect records within that range.

4. Product Backlog Prioritization

Priority is based on essential user value, dependencies and risk. MoSCoW-style labels are used: Must Have, Should Have and Could Have.

Rank	Story	Priority	Reason / Dependency
1	US-01 Lost report	Must Have	Core entry point; required for the lost-item workflow.
2	US-02 Found report	Must Have	Core entry point; matching requires found-item data.
3	US-03 Search/filter	Must Have	Makes saved reports discoverable; depends on stored reports.
4	US-06 Verify and record handover	Must Have	Important safety control for secure recovery.
5	US-04 Possible matching	Should Have	Improves discovery; depends on lost and found reports.
6	US-05 Notifications	Should Have	Makes users aware of possible matches; depends on matching and contact settings.
7	US-07 Moderation	Should Have	Supports quality and safe administration.
8	US-08 Analytics	Could Have	Useful for management, but not required for the initial reporting workflow.

Dependencies and Release Logic

- Reporting stories (US-01 and US-02) must create reliable records before search and matching can work.
- Search (US-03) can be delivered after report storage is implemented.
- Matching (US-04) precedes match notifications (US-05).
- Verification and handover (US-06) should be available before the platform is used for real item returns.

5. Sprint Backlog and Story Point Estimation

Sprint 1 Plan

Sprint duration: 2 weeks. Team assumption: a small student team with a product owner, developer(s) and tester. Sprint 1 focuses on a usable reporting and search foundation; advanced matching and notifications are planned for a later sprint.

Sprint Goal

Enable a user to create a lost-item or found-item report and search existing listings, with required-field validation and saved report records.

Selected Story	Points	Sprint Tasks	Task Output
US-01 Lost report	5	Design report fields and form; implement validation; save report and reference ID; test valid and invalid submissions.	Working lost-report form and saved records.
US-02 Found report	5	Implement found-item fields; support optional image upload; validate file type/size; test report submission.	Working found-report form with optional photograph.
US-03 Search/filter	3	Build listing page; add keyword/category filters; show empty state; test matching and non-matching searches.	Searchable lost/found listings.

Sprint 1 commitment: 13 story points. This is a planning estimate, not a measured team velocity. The team should adjust the commitment after considering its actual availability and skills.

Story Point Estimation and Justification

Story	Points	Justification
US-01	5	Medium effort: form design, validation, persistence and several test cases.
US-02	5	Medium effort: similar report workflow plus optional image upload and file validation.
US-03	3	Small-to-medium effort: listing and basic filters; depends on report records being available.
US-04	5 (later)	Matching rules require choosing fields, scoring candidates and testing false matches.
US-05	3 (later)	Notification workflow depends on matching and a configured delivery mechanism.
US-06	5 (later)	Includes role checks, verification outcome, handover record and status changes.

Technique: Planning Poker with a Fibonacci-style scale (1, 2, 3, 5, 8, 13). Team members compare complexity, uncertainty and testing effort, discuss large differences, and agree on one estimate. Story points represent relative effort, not hours.

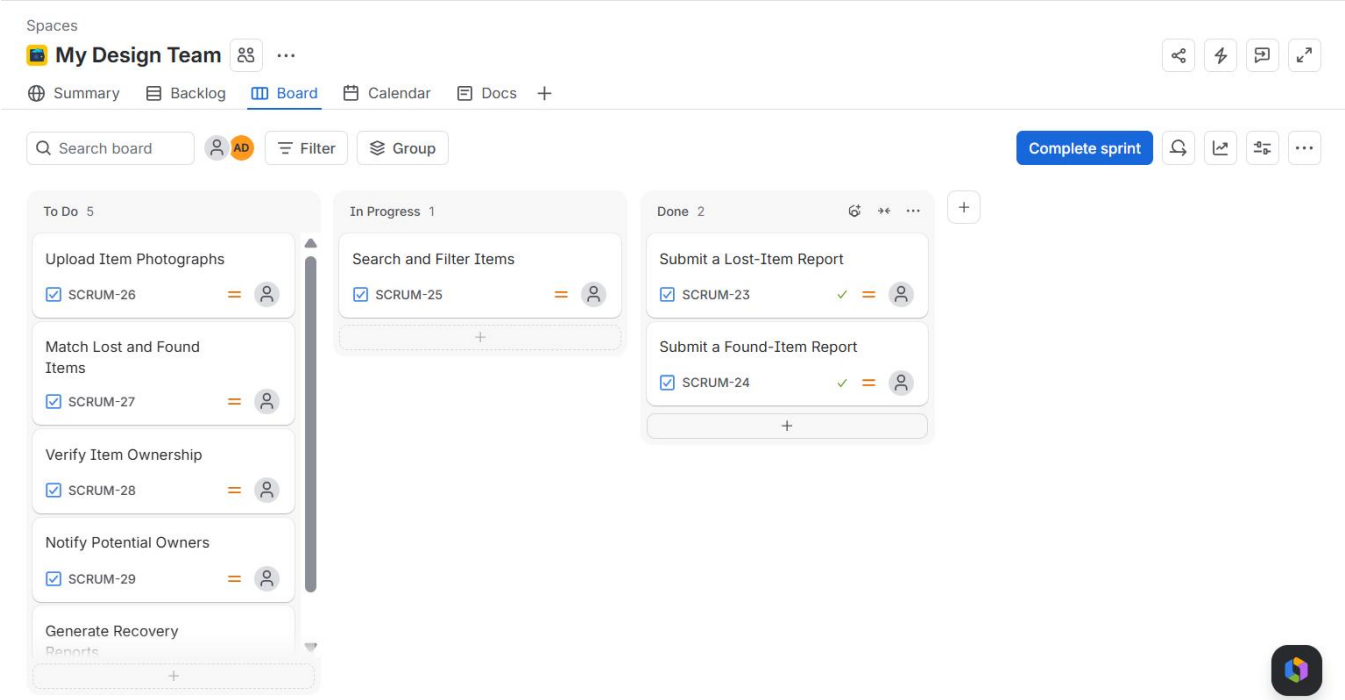
Expected Deliverables and Review

- A working prototype with lost and found report forms.
- Validated required fields and stored report records with reference identifiers.
- A listing page with keyword/category search and a useful empty-results message.
- Test evidence for valid submission, missing fields, optional image handling and search results.
- Sprint review demonstration, stakeholder feedback and an updated backlog for Sprint 2.

Risks and Mitigation

- False matches: label suggestions as possible matches and require human ownership verification.
- Sensitive details: restrict access to contact information and administrative actions.
- Scope growth: keep matching, notifications and analytics outside Sprint 1 unless core stories are complete.

JIRA SCRUM



Spaces

My Design Team

Summary | Backlog | Board | Calendar | Docs

Search ba... AD Filter

SCRUM Sprint 2 20 Oct – 3 Nov (8 work items) 0 0 0 Complete sprint

~~SCRUM-23~~	Submit a Lost-Item Report	Done	-
~~SCRUM-24~~	Submit a Found-Item Report	Done	-
SCRUM-25	Search and Filter Items	In Progress	-
SCRUM-26	Upload Item Photographs	To Do	-
SCRUM-27	Match Lost and Found Items	To Do	-
SCRUM-28	Verify Item Ownership	To Do	-
SCRUM-29	Notify Potential Owners	To Do	-
SCRUM-30	Generate Recovery Reports	To Do	-

+ Create